

Umberto Villa, Ph.D.

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My research interests and expertise are in computational mathematics and imaging science. My work is uniquely informed by my transdisciplinary training and approaches, combining mathematical and statistical modeling with engineering and computational sciences. My ultimate goal is to exploit the power of scientific computing and artificial intelligence to accelerate scientific discovery and engineering innovation. My current research focuses on advancing emerging quantitative image modalities to help resolve major challenges in medicine and public health, including early detection of cancers and improved treatment outcomes.

EDUCATION	Emory University , Atlanta, GA (United States)	
	<i>PhD in Mathematics</i>	2012
	Politecnico di Milano , Milan (Italy) & Politecnico di Torino , Turin (Italy)	
	<i>ASP diploma - Alta Scuola Politecnica</i>	2008
	Politecnico di Milano , Milan (Italy) & Politecnico di Torino , Turin (Italy)	
	<i>Dual Master's degree in Mathematical Engineering, cum laude</i>	2007
	Politecnico di Milano , Milan (Italy)	
	<i>Bachelor's degree in Mathematical Engineering, cum laude</i>	2005
EMPLOYMENT	The University of Texas at Austin , Austin, TX	
	<i>Assistant Professor</i> , Department of Biomedical Engineering	2025 –
	<i>Core faculty</i> , Oden Institute for Computational Engineering and Science	2025 –
	<i>Research Associate Professor</i>	2024 – 2025
	<i>Research Scientist</i>	2022 – 2024
	<i>Research Associate</i>	2015 – 2018
	University of Illinois , Urbana-Champaign, IL	
	Department of Bioengineering	
	<i>Adjunct Research Associate Professor</i>	2024 –
	<i>Adjunct Research Assistant Professor</i>	2020 – 2024
	Washington University in St. Louis , St. Louis, MO	
	Electrical & Systems Engineering	
	<i>Research Assistant Professor</i>	2018 – 2022
	<i>Imaging Science Ph.D. Program Faculty</i>	2018 – 2022
	<i>Institute of Public Health Faculty Scholar</i>	2020 – 2022
	Lawrence Livermore National Laboratory (LLNL) , Livermore, CA	
	Center for Applied Scientific Computing	
	<i>Visiting Scientist</i>	2015 – 2021
	<i>Postdoctoral Researcher</i>	2013 – 2015
	<i>Student Internship</i>	Summers 2011 & 2012
	Oak Ridge National Laboratory (ORNL) , Oak Ridge, TN	
	Computer Science and Mathematics Division	
	<i>Student Internship</i>	Summers 2009 & 2010

HONORS,
FELLOWSHIPS
AND
AWARDS

1. Best Student Paper Award, 12th Copper Mountain Conference on Iterative Methods, Copper Mountain, Colorado, US **2012**
2. Laney Graduate School Scholarship, Emory University, Atlanta, GA **2008 – 2012**
3. Alta Scuola Politecnica Scholarship, Politecnico of Milano, Milan, Italy, **2005 – 2007**
4. Medal for best graduate recipient, B.S. in Mathematical Engineering, Politecnico of Milano, Milan, Italy **2005**
5. International Mathematical Olympiad (National phase), Cesenatico, Italy **2001**

Honors and awards as part of a team

6. Seno Medical Best Paper Award, Photons Plus Ultrasound: Imaging and Sensing 2022, SPIE Photonics West BIOS, San Francisco, CA, US **2022**

Honors and awards to students and mentees:

7. Hsuan-Kai (Kevin) Huang (PhD candidate at UIUC I co-advise): Seno Medical Best Paper Award, Photons Plus Ultrasound: Imaging & Sensing 2026, SPIE Photonics West BIOS, San Francisco, CA, US **2026**
8. Thomas Wynn (Undergraduate student that I mentored): Fulbright Fellowship **2025**
9. Graham Pash (PhD candidate for which I serve as committee member): Bavarian Graduate School of Computational Engineering (BGCE) Student Paper Price, 2025 SIAM Conference on Computational Sciences & Engineering **2025**
10. Simone Puel (PhD candidate I served as informal mentor for): Geophysical Journal International (GJI) 2024 Student Paper Award. **2024**
11. Luke Lozenski (PhD candidate I advise): Imaging Science Pathway Fellowship, Washington University in St Louis. Full salary support. **2022– 2024**
This fellowship, funded by an NIH T32 Training Grant, is a highly competitive award that recognize academic excellence of graduate students in the McKelvey School of Engineering and the Medical School at Washington University.
12. Refik Mert Cam (PhD student I co-advise): Elsa and Floyd Dunn award **2024**
This award honors graduate students with demonstrated research interest in topics including, but not limited to, biomedical ultrasound, bioengineering, and related fields.
13. Fu Li (PhD student I co-advise): Ultrasonic Imaging and Tomography Cum Laude Poster Award, SPIE Conference on Medical Imaging, **2024**
14. Luke Lozenski (PhD candidate I advise): Applied Machine Learning Summer Fellowship, Los Alamos National Laboratory, **2023**

GRANTS AND
CONTRACTS

Awarded research grants

1. **U. Villa (PI)**, M Anastasio, S Ermilov, M Pagel (Co-Is). *Advancing three-dimensional pre-clinical dynamic contrast-enhanced photoacoustic computed tomography via quantitative image reconstruction*, National Institutes of Health (NIH), National Institute of Biomedical Imaging and Bioengineering, NIH R01 EB034261
02/15/24–01/31/28 **\$2,352,889**
2. Lihong Wang (PI); M. Anastasio (Co-I), **U. Villa (Co-I & subaward PI)**. *Deepening optical imaging with next-generation photoacoustic computed tomography*, National Institutes of Health (NIH), National Institute on Drug Abuse, NIH UG3 DA065155
09/01/2025–08/31/2027 **\$3,179,146**
Subaward amount \$56,142
3. **U. Villa**, M. Pagel (**MPIs**), *Safe, Accurate Assessment of Treatment Response via Dynamic Contrast Enhanced Multispectral Optoacoustic Tomography Imaging of Tumor Perfusion*, The Joint Center for Computational Oncology (JCCO), Oncological Data and Computational Sciences program.
09/01/23 – 08/31/24 **\$50,000** (Direct Costs)

4. M. Anastasio, A. Oraevsky (MPIs); F. Brooks (Co-I), **U. Villa (Co-I & subaward PI)**, *A Computational Framework Enabling Virtual Imaging Trials of 3D Quantitative Optoacoustic Tomography Breast Imaging*, National Institutes of Health (NIH), National Institute of Biomedical Imaging and Bioengineering, NIH R01EB031585
08/01/22 – 04/30/26 **\$2,308,021**
Subaward amount \$391,984
5. S. Ermilov (PI), M. Anastasio (Co-I), S. Emelianov (Co-I), **U. Villa (Co-I & subaward PI)**, *Integrated photoacoustic and fluorescence imaging system for anatomical, functional, and molecular characterization of murine models*, National Institutes of Health (NIH), Novel Tools and Devices for Animal Research Facilities and to Support Care of Animal Models, NIH R44OD023029
08/15/22—07/31/24 **\$1,924,204**
Subaward amount \$96,068
6. M. Anastasio and N. Duric (MPIs); **U. Villa (Co-I & subaward PI)**, *Advanced image reconstruction for accurate and high-resolution breast ultrasound tomography*, National Institutes of Health, National Institute of Biomedical Imaging and Bioengineering, NIH R01EB028652
09/01/19—08/31/24 **\$2,179,420**
Subaward amount \$250,761
7. T. Kim (PI); L. Henke, G. Hugo, C. Park, M. Schmidt, **U. Villa**, H. Yi (**Co-Is**), *MRI augmented X-ray imaging-guided adaptive radiotherapy for pancreatic cancer (MAX-guided ART)*, Siteman Investment Program (Pre-R01 Award)
07/01/21–06/30/23 **\$200,000**
8. **U. Villa (PI)**, *ADLA: Automatic differentiation and local assembly of exotic finite element variational forms in MFEM*, LLNL B638337 subcontract
11/01/19–10/31/20 **\$59,999**
9. O. Ghattas (PI) and **U. Villa (Co-PI)**, *Collaborative Research: SI2-SSI: Integrating Data with Complex Predictive Models under Uncertainty: An Extensible Software Framework for Large-Scale Bayesian Inversion*, National Science Foundation, Division of Advanced Cyberinfrastructure, Grant ACI-1550593
09/01/16–08/31/20 **\$350,885**
A collaborative research project (separate awards) with N. Petra (UC-Merced), Y. Marzouk and M. Parno (MIT) with total funding of \$1.35M

In-Kind research grants: Computational resources

10. **U. Villa**, *Advancing biomedical sciences via virtual imaging trials and AI*, TACC allocation ASC25089, 25,000 GPU-node hours, 2025–2026
11. **U. Villa**, M. Pagel (**MPI**), *Safe, Accurate Assessment of Treatment Response via Dynamic Contrast Enhanced Multispectral Optoacoustic Tomography Imaging of Tumor Perfusion*, The Joint Center for Computational Oncology (JCCO), Oncological Data and Computational Sciences program, 12,500 SUs, 2023–2024
12. M. Anastasio (PI), **U. Villa (Co-PI)**, *Distributed GPU-accelerated image reconstruction methods for breast ultrasound computed tomography*, Illinois Delta research allocation, 16,000 GPU-hours, 2022
13. M. Anastasio (PI), **U. Villa (Co-PI)**, *A computational framework integrating wave physics simulation and machine learning for fast and accurate transcranial photoacoustic tomography reconstruction*, Illinois Blue Waters research allocation, 210,000 node-hours, 2021
14. M. Anastasio (PI); J. Poudel, **U. Villa (Co-PI)**, *Safe and rapid functional brain imaging with transcranial photoacoustic tomography: Accelerating iterative image reconstruction algorithms using GPUs*, Illinois Blue Waters research allocation, 210,000 node-hours (estimated value of awarded resources \$130,263), 2020

Awarded educational grants

15. O. Ghattas, Y. Marzouk, M. Parno, N. Petra, G. Stadler, and **U. Villa**, *2018 Gene Golub SIAM Summer School entitled Inverse Problems: Systematic Integration of*

In-Kind educational grants: Computational resources

16. **U. Villa (PI)**, Computing resources for the graduate level course *Tools and Techniques of Computational Science*, TACC Instructional allocation CSE-380-Tools-and-Te, 2022–2024
17. **U. Villa (PI)**, Computing resources for the graduate level course *Computational and Variational Inverse Problems*, Explore ACCESS (educational) allocation MTH230002, 400,000 credits, 2023–2024
18. **U. Villa (PI)**, Cloud computing resources for the graduate level course on *Computational Methods in Imaging Science*, XSEDE educational allocation TG-SEE190001, 100,000 CPU hours (estimated value of awarded resources \$8,445), 2019–2020
19. **U. Villa (PI)** and N. Petra (Co-PI), Cloud computing resources for the *2018 Gene Golub SIAM Summer School* entitled *Inverse Problems: Systematic Integration of Data with Models under Uncertainty*, XSEDE educational allocation TG-DMS180009, 60,000 CPU hours (estimated value of awarded resources \$10,014), 2018.

RESEARCH EXPERIENCE

Advancing three-dimensional preclinical dynamic contrast-enhanced photoacoustic computed tomography via quantitative image reconstruction

The broad objective of this collaborative project is to develop, refine, and validate 4D and 5D image reconstruction methods with applications to dynamic contrast enhanced photoacoustic computed tomography. The proposed approach leverages non-convex optimization techniques and proximal methods to incorporate data-driven and low-rank based image priors.

Funding: NIH, National Institute of Biomedical Imaging and Bioengineering (NIBIB), R01 EB034261.

Role: **PI** **2024 –**

Exascale Predictive Simulation of Inductively Coupled Plasma Torches

The primary objective of the center is to develop an advanced integrated predictive computational model for an inductively-coupled plasma (ICP) torch, which can make effective use of emerging exascale computing hardware. Funding: Department of Energy; Advanced Simulation and Computing Predictive Science Academic Alliance Program (PSAAP III)

Role: **Senior Software Architect** **2023 –**

A Computational Framework Enabling Virtual Imaging Trials of 3D Quantitative Optoacoustic Tomography Breast Imaging

This project addresses the challenges of reducing over-diagnosis and over-treatment of breast cancer by developing transformative computational methods (learned and/or model-based) to enable three-dimensional (3D) quantitative optoacoustic tomography (OAT) of the vasculature and oxygen saturation distribution within the human breast.

Funding: NIH, National Institute of Biomedical Imaging and Bioengineering (NIBIB), R01EB031585.

M. Anastasio, A. Oraevsky (PI)

Role: **Co-Investigator/Subaward PI** **2022 –**

Safe, Accurate Assessment of Treatment Response via Dynamic Contrast Enhanced Multispectral Optoacoustic Tomography Imaging of Tumor Perfusion

The broad objective of this collaborative project with Dr. Pagel (MD Anderson) is develop novel and advanced computational and mathematical models to accurately estimate tumor perfusion rates from DCE MSOT images. The proposed methods will be assessed using retrospective DCE MSOT imaging data from a large cohort of 120 mice undergoing radiotherapy (three tumor models, four doses tested).

Funding: The Joint Center for Computational Oncology (JCCO), Oncological Data and Computational Sciences program

Role: **PI** (MD Anderson PI: Marty Pagel) **2023 – 2024**

Integrated photoacoustic and fluorescence imaging system for anatomical, functional, and molecular characterization of murine models

This project aims at developing learning-enhanced image reconstruction to enable quantitative dynamic PACT imaging from a reduced number of tomographic views.

Funding: NIH, Small Business Innovation Research Grants (SBIR) , R44OD023029. S. Ermilov (PI)
Role: **Co-Investigator/Subaward PI** **2022 – 2024**

Advanced image reconstruction for accurate and high-resolution breast ultrasound tomography

The broad objective of this project is to maximize the clinical utility of ultrasound computed tomography (USCT) for whole breast imaging by significantly advancing the state-of-the-art in USCT image reconstruction using model-based and learning methods.

Funding: NIH, National Institute of Biomedical Imaging and Bioengineering (NIBIB), R01EB028652. M. Anastasio, N. Duric (PIs)

Role: **Co-Investigator/Subaward PI** **2018 – 2023**

Integrating Data with Complex Predictive Models under Uncertainty: An Extensible Software Framework for Large-Scale Bayesian Inversion

The goal of this study is to develop, disseminate, and support a robust, scalable, high-performance, open-source software framework incorporating a suite of advanced Bayesian inversion algorithms.

Funding: NSF, Office of Advanced Cyberinfrastructure, ACI-1550593. O. Ghattas (PI)

Role: **Co-PI** **2016 – 2020**

Large-scale Inverse Problems and Uncertainty Quantification for Reservoir Modeling

The focus of this joint ExxonMobil-UTCI project is to develop state-of-the-art inversion and uncertainty quantification methods to reservoir models with complex features including faults.

Funding: Joint ExxonMobil-UT Energy Institute Project, UTA17-000408 (EM10480.14). O. Ghattas (PI), G. Biros, T. Bui-Thanh, C. Dawson (Co-PIs)

Role: Research scientist **2017 – 2019**

Bayesian Optimal Experimental Design for Inverse Scattering

The goal of this study is to develop a rigorous Bayesian framework to design source/receiver configuration to maximize identifiability.

Funding: AFOSR, Computational Mathematics program, FA9550-17-1-0190. O. Ghattas (PI), G. Biros and Y. Marzouk (Co-PIs)

Role: Research scientist **2017 – 2018**

Inference, Simulation, and Optimization of Complex Systems Under Uncertainty: Theory, Algorithms, and Applications to Turbulent Combustion

This project developed an end-to-end, integrated uncertainty quantification framework enabling us to quantify, manage, and minimize uncertainty in large scale multiscale/multiphysics problems.

Funding: DARPA, EQUIPS program, W911NF-15-2-0121. O. Ghattas (PI), R. Moser, G. Biros, K. Willcox, M. Heinkenschloss, A. Stuart, M. Girolami, A. Philpott (Co-PIs)

Role: Research scientist **2016 – 2017**

Towards Optimal Order Resilient Solvers at Extreme Scale (TOORSES)

This project developed large scale linear solvers and preconditioners exploiting multilevel techniques and hierarchical matrices factorizations.

Funding: DOE Office of Advanced Scientific Computing Research. X.-S. Li (lead PI), P. Vassilevski (LLNL PI)

Role: Postdoctoral researcher **2013 – 2015**

Scalable Multilevel UQ Concepts for Extreme-Scale Multiscale Problems

The objective of this project is to develop multilevel techniques to accelerate forward and inverse uncertainty quantification (UQ) tasks involving complex multiphysics partial differential equations models.

Funding: DOE Office of Advanced Scientific Computing Research. Y. Efendiev (lead PI), P. Vassilevski (LLNL PI)

Role: Postdoctoral researcher **2013 – 2015**

Adaptive Dimension Reduction via Coarsening and Multilevel Solvers

This project investigates highly efficient mathematical tools to construct coarse spaces and respective coarse models that are operator-dependent and to expand the applicability of multigrid methods to very general partial differential equations, such as mixed formulations and saddle point systems.

Funding: DOE Office of Advanced Scientific Computing Research. P. Vassilevski (PI)

Role: Graduate research assistant **2011 – 2012**

Multiphysics Multimodel Domain Decomposition: an Application to Conjugate Heat Transfer

This project investigates a general optimization-based framework for multiphysics multimodel Domain Decomposition with applications to conjugate heat transfer and fluid structure interaction problems.

Funding: ORNL Laboratory Directed Research and Development (LDRD). J. Hill (PI)

Role: Graduate research assistant **2009 – 2010**

Scalable Efficient Methods for Incompressible Fluid-dynamics in Engineering Problems (PhD thesis)
 Analysis and implementation of a new time-adaptive algorithm for the solution of the unsteady Navier-Stokes equations.

Development of parallel and scalable block preconditioners for saddle point problems.

Application of these new numerical methods to patient specific blood flow simulations with the aim to numerically investigate pathological or clinical flow conditions (e.g. formation of aneurysms in the carotid artery, design of left ventricle assisting devices).

Advisor: Dr. Alessandro Veneziani

2008 – 2012

PUBLICATIONS

Complete list of Peer-Reviewed Journal Articles

Scientific journal articles as *first* or *senior* author:

- [1] Scope Crafts, E., **Villa, U.**, “Benchmarking Diffusion Annealing-Based Bayesian Inverse Problem Solvers”. In: *IEEE Open Journal of Signal Processing* 6 (2025), pp. 975–991. DOI: 10.1109/OJSP.2025.3597867.
- [2] Craft Scope, E., Anastasio, M. A., **Villa, U.**, “Optimizing Quantitative Photoacoustic Imaging Systems: The Bayesian Cramér-Rao Bound Approach”. In: *Inverse Problems* 40.12 (2024), p. 125012. DOI: 10.1088/1361-6420/ad910a.
- [3] Lozenski, L., Cam, R. M., Pagel, M. D., Anastasio, M. A., **Villa, U.**, “ProxNF: Neural Field Proximal Training for High-Resolution 4D Dynamic Image Reconstruction”. In: *Transactions on Computational Imaging* 10 (2024), pp. 1368–1383. DOI: 10.1109/TCI.2024.3458397.
- [4] Cam, R. M., Wang, C., Thompson, W., Ermilov, S. A., Anastasio, M. A., **Villa, U.**, “Spatiotemporal Image Reconstruction to Enable High-Frame Rate Dynamic Photoacoustic Tomography with Rotating-Gantry Volumetric Imagers”. In: *Journal of Biomedical Optics* 29.S1 (2024), S11516. DOI: 10.1117/1.JBO.29.S1.S11516.
- [5] Lozenski, L., Wang, H., Li, F., Anastasio, M., Wohlberg, B., Lin, Y., **Villa, U.**, “Learned Full Waveform Inversion Incorporating Task Information for Ultrasound Computed Tomography”. In: *IEEE Transactions on Computational Imaging* 10 (2024), pp. 69–82. DOI: 10.1109/TCI.2024.3351529.
- [6] Kalchev, D. Z., Vassilevski, P. S., **Villa, U.**, “Parallel Element-based Algebraic Multigrid for $H(\text{curl})$ and $H(\text{div})$ Problems Using the ParELAG Library”. In: *SIAM Journal on Scientific Computing* 45.3 (2023), S371–S400. DOI: 10.1137/21M1433253.
- [7] Lozenski, L., Anastasio, M. A., **Villa, U.**, “A Memory-Efficient Self-Supervised Dynamic Image Reconstruction Method Using Neural Fields”. In: *IEEE Transactions on Computational Imaging* 8 (2022), pp. 879–892. DOI: 10.1109/TCI.2022.3208511.
- [8] **Villa, U.**, Petra, N., Ghattas, O., “hIPPYlib: An Extensible Software Framework for Large-Scale Inverse Problems Governed by PDEs; Part I: Deterministic Inversion and Linearized Bayesian Inference”. In: *ACM Trans. Math. Softw.* 47.2 (Apr. 2021). ISSN: 0098-3500. DOI: 10.1145/3428447.
- [9] **Villa, U.**, Petra, N., Ghattas, O., “hIPPYlib: an Extensible Software Framework for Large-scale Deterministic and Bayesian Inverse Problems”. In: *Journal of Open Source Software* 3.30 (2018), p. 940. DOI: 10.21105/joss.00940.
- [10] Christensen, M., Vassilevski, P. S., **Villa, U.**, “Nonlinear Multigrid solvers exploiting AMGe coarse spaces with approximation properties”. In: *Journal of Computational and Applied Mathematics* 340 (2018), pp. 691–708. ISSN: 0377-0427. DOI: 10.1016/j.cam.2017.10.029.
- [11] Osborn, S., Vassilevski, P. S., **Villa, U.**, “A Multilevel Hierarchical Sampling Technique for Spatially Correlated Random Fields”. In: *SIAM Journal on Scientific Computing* 39.5 (2017), S543–S562. DOI: 10.1137/16M1082688. eprint: 1703.08498.
- [12] Vassilevski, P. S., **Villa, U.**, “A mixed formulation for the Brinkman problem”. In: *SIAM Journal on Numerical Analysis* 52.1 (2014), pp. 258–281. DOI: 10.1137/120884109.
- [13] Vassilevski, P. S., **Villa, U.**, “A block-diagonal algebraic multigrid preconditioner for the Brinkman problem”. In: *SIAM Journal on Scientific Computing* 35.5 (2013), S3–S17. DOI: 10.1137/120882846.
- [14] Veneziani, A., **Villa, U.**, “ALADINS: An ALgebraic splitting time ADaptive solver for the Incompressible Navier–Stokes equations”. In: *Journal of Computational Physics* 238 (2013), pp. 359–375. DOI: 10.1016/j.jcp.2012.11.049.

Publications with significant contributions in all the research aspects (refinement of the original idea, study design, implementation, writing, mentoring of students/trainees)

- [15] Huang, H.-K., Kuo, J., Park, S., **Villa, U.**, Wang, L. V., Anastasio, M. A., “Stochastic numerical head phantoms to enable virtual imaging studies of transcranial photoacoustic computed tomography”. In: *Photoacoustics* (2026), p. 100859. DOI: 10.1016/j.pacs.2026.100859.
- [16] Yang, K., Park, S., Jeong, G., Huang, H.-K., Oraevsky, A. A., **Villa, U.**, Anastasio, M. A., “A Learning-based Framework for Spatial Impulse Response Compensation in 3D Photoacoustic Computed Tomography”. In: *IEEE Transactions on Medical Imaging* (2026). DOI: 10.1109/TMI.2026.3717230.
- [17] Nicholson, R., Vuchkov, R., **Villa, U.**, Petra, N., “Taylor approximation variance reduction for approximation errors in PDE-constrained Bayesian inverse problems”. In: *Inverse Problems* 42.4 (Apr. 2026), p. 045020. DOI: 10.1088/1361-6420/ae609b.
- [18] Park, S., Jeong, G., **Villa, U.**, Anastasio, M. A., “A virtual imaging framework for three-dimensional quantitative optoacoustic tomography using stochastic numerical Breast phantoms”. In: *Photoacoustics* (June 2026), p. 100850. ISSN: 2213-5979. DOI: 10.1016/j.pacs.2026.100850.
- [19] Lin, Y., Feng, S., Theiler, J., Chen, Y., **Villa, U.**, Rao, J., Greenhall, J., Pantea, C., Anastasio, M. A., Wohlberg, B., “Survey of Deep Learning and Physics-Based Approaches in Computational Wave Imaging”. In: *Proceedings of the IEEE* 113.12 (2025), pp. 1384–1417. DOI: 10.1109/JPROC.2026.3684003.
- [20] Cam, R. M., Park, S., **Villa, U.**, Anastasio, M. A., “Application of a virtual imaging framework for investigating a deep learning-based reconstruction method for 3D quantitative photoacoustic computed tomography”. In: *Photoacoustics* 48 (2026), p. 100792. ISSN: 2213-5979. DOI: <https://doi.org/10.1016/j.pacs.2025.100792>.
- [21] Pash, G., **Villa, U.**, Hormuth, D. A. I., Yankeelov, T. E., Willcox, K., “Predictive Digital Twins with Quantified Uncertainty for Patient-Specific Decision Making in Oncology”. In: Available on-line (2026). DOI: 10.1016/j.jcp.2026.114937.
- [22] Hong, T., **Villa, U.**, Fessler, J. A., “A Convergent Generalized Krylov Subspace Method for Compressed Sensing MRI Reconstruction with Gradient-Driven Denoisers”. In: *IEEE Transactions on Computational Imaging* (2026), pp. 1–14. DOI: 10.1109/TCI.2026.3655489.
- [23] Chen, P., Park, S., Cam, R. M., Huang, H.-K., Oraevsky, A. A., **Villa, U.**, Anastasio, M. A., “Learning a Filtered Backprojection Reconstruction Method for Photoacoustic Computed Tomography with Hemispherical Measurement Geometries”. In: *IEEE Trans on Medical Imaging* Early Access (2025), p. 13. DOI: 10.1109/TMI.2025.3591706.
- [24] Jeong, G., **Villa, U.**, Anastasio, M., “Revisiting the joint estimation of initial pressure and speed-of-sound distributions in photoacoustic computed tomography with consideration of canonical object constraints”. In: *Photoacoustics* 43 (2025), p. 100700. DOI: 10.1016/j.pacs.2025.100700.
- [25] Huang, H.-K., Kuo, J., Zhang, Y., Aborahama, Y., Cui, M., Sastry, K., Park, S., **Villa, U.**, Wang, L. V., Anastasio, M., “Fast aberration correction in 3D transcranial photoacoustic computed tomography via a learning-based image reconstruction method”. In: *Photoacoustics* 43 (2025), p. 100698. DOI: 10.1016/j.pacs.2025.100698.
- [26] Li, K., **Villa, U.**, Li, H., Anastasio, M., “Application of Learned Ideal Observers for Estimating Task-Based Performance Bounds for Computed Imaging Systems”. In: *Journal of Medical Imaging* 11.2 (2024), p. 026002. DOI: 10.1117/1.JMI.11.2.026002.
- [27] Jeong, G., Li, T., Duric, N., **Villa, U.**, Anastasio, M., “Investigating the Use of Traveltime and Reflection Tomography for Deep Learning-Based Sound-Speed Estimation in Ultrasound Computed Tomography”. In: *IEEE Trans on Ultrasonic, Ferroelectrics, and Frequency Control* 71.11 (2024), pp. 1358–1376. DOI: 10.1109/TUFFC.2024.3459391.
- [28] Cam, R. M., **Villa, U.**, Anastasio, M. A., “Learning a Stable Approximation of an Existing but Unknown Inverse Mapping: Application to the Half-Time Circular Radon Transform”. In: *Inverse Problems* 40.8 (2024), p. 085002. DOI: 10.1088/1361-6420/ad4f0a.
- [29] Kim, K.-T., **Villa, U.**, Parno, M., Marzouk, Y., Ghattas, O., Petra, N., “hIPPYlib-MUQ: A Bayesian Inference Software Framework for Integration of Data with Complex Predictive Models under Uncertainty”. In: *ACM Trans. Math. Softw.* 49.2 (2023). DOI: 10.1145/3580278.
- [30] Puel, S., Becker, T. W., **Villa, U.**, Ghattas, O., Liu, D., “An adjoint-based optimization method for jointly inverting heterogeneous material properties and fault slip from earthquake surface deformation data”. In: *Geophysical Journal International* 236 (2 2023), pp. 778–797. DOI: 10.1093/gji/ggad442.
- [31] O’Leary-Roseberry, T., Chen, P., **Villa, U.**, Ghattas, O., “Derivative Informed Neural Operator: An Efficient Framework for High-Dimensional Parametric Derivative Learning”. In:

- Journal of Computational Physics* (2023), p. 112555. ISSN: 0021-9991. DOI: 10.1016/j.jcp.2023.112555.
- [32] Li, F., **Villa, U.**, Duric, N., Anastasio, M. A., “A forward model incorporating elevation-focused transducer properties for 3D full-waveform inversion in ultrasound computed tomography”. In: *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* 70.10 (2023), pp. 1339–1354. DOI: 10.1109/TUFFC.2023.3313549.
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 - [111] Li, F., **Villa, U.**, Park, S., He, S., Anastasio, M. A., “A framework for ultrasound computed tomography virtual imaging trials that employs anatomically realistic numerical breast phantoms”. In: *Medical Imaging 2021: Ultrasonic Imaging and Tomography*. Ed. by Brett C. Byram and Nicole V. Ruiter. Vol. 11602. International Society for Optics and Photonics. SPIE, 2021. DOI: 10.1117/12.2582260.
 - [112] Wang, C., **Villa, U.**, Thompson, W., Park, S., Ermilov, S. A., Anastasio, M. A., “Dynamic reconstruction of three-dimensional photoacoustic tomography from few projections”. In: *Photons Plus Ultrasound: Imaging and Sensing 2021*. Ed. by Alexander A. Oraevsky and Lihong V. Wang. Vol. 11642. International Society for Optics and Photonics. SPIE, 2021. DOI: 10.1117/12.2585344.
 - [113] Park, S., **Villa, U.**, Brooks, F. J., Su, R., Oraevsky, A. A., Anastasio, M. A., “Three-dimensional quantitative functional optoacoustic tomography to estimate vascular blood oxygenation of the breast”. In: *Photons Plus Ultrasound: Imaging and Sensing 2021*. Ed. by Alexander A. Oraevsky and Lihong V. Wang. Vol. 11642. International Society for Optics and Photonics. SPIE, 2021. DOI: 10.1117/12.2585341.
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 - [119] Christensen, M., **Villa, U.**, Vassilevski, P. S., “Multilevel techniques lead to accurate numerical upscaling and scalable robust solvers for reservoir simulation”. In: *SPE Reservoir Simulation Symposium*. 23-25 February, Houston, Texas, USA, SPE-173257-MS. Society of Petroleum Engineers. 2015.

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- [121] Slawinski, J., **Villa, U.**, Passerini, T., Veneziani, A., Sunderam, V., “Issues in Communication Heterogeneity for Message-Passing Concurrent Computing”. In: *27th IEEE Intl. Parallel and Distributed Processing Symposium Workshops & PhD Forum (IPDPSW)*. IEEE. 2013, pp. 93–102. DOI: 10.1109/IPDPSW.2013.140.
- [122] Slawinski, J., **Villa, U.**, Passerini, T., Veneziani, A., Sunderam, V., “Experiences with Target-Platform Heterogeneity in Clouds, Grids, and On-Premises Resources”. In: *26th IEEE Intl. Parallel and Distributed Processing Symposium Workshops & PhD Forum (IPDPSW)*. IEEE. 2012, pp. 41–52. DOI: 10.1109/IPDPSW.2012.20.

Theses and dissertations

- [123] **Villa, U.** “Scalable Efficient Methods for Incompressible Fluid-dynamics in Engineering Problems”. Advisor: A. Veneziani. PhD thesis. Atlanta, GA: Emory University, 2012.
- [124] **Villa, U.** “Environment & energy - Hydrogen: opportunities and utilization”. Advisors: F. Profumo, E. Paolucci, A. Tenconi. External Institutions: Centro Estero Camere di Commercio Piemontesi, STEP Ricerche S.r.l. Alta Scuola Politecnica diploma. Politecnico di Milano & Politecnico di Torino, 2008.
- [125] **Villa, U.** “Finite Element Analysis of the Brake Pad System and Multibody Modeling of Motor Vehicles in Braking-Phase”. Advisor: A. Veneziani; Committee: L. Trainelli, A. Vigliani. MA thesis. Milan, Italy: Politecnico di Milano, 2007.
- [126] Vele, S., **Villa, U.**, “Mathematical modeling and numerical simulation of hemodynamics problems in one dimension”. Advisor: A. Veneziani. Bachelor’s thesis. Milan, Italy: Politecnico di Milano, 2005.

OTHER RESEARCH PRODUCTS

Scientific datasets

- [127] Park, S., Jeong, G., **Villa, U.**, Anastasio, M., *Stochastic Optoacoustic Numerical Breast Phantoms with Multiple Skin Tones and Simulated OAT Measurement Data*. 2026. DOI: 10.13012/B2IDB-1071666_V1.
- [128] Huang, H.-K., Kuo, J., Zhou, B., Park, S., **Villa, U.**, Anastasio, M., *Stochastic Photoacoustic Numerical Head Phantoms and Simulated PACT Measurement Data*. 2026. DOI: 10.13012/B2IDB-5613691_V1.
- [129] Park, S., Jeong, G., **Villa, U.**, Anastasio, M., *Anatomy-Matched Multi-Variant Stochastic Optoacoustic Numerical Breast Phantoms and Simulated OAT Measurement Data*. 2026. DOI: 10.13012/B2IDB-8164905_V1.
- [130] Jeong, G., **Villa, U.**, Park, S., Anastasio, M. A., *2D Acoustic Numerical Breast Phantoms and Measurement Data for Joint Reconstruction in PACT*. 2025. DOI: 10.13012/B2IDB-9572729_V1.
- [131] Li, F., **Villa, U.**, Park, S., Jeong, G., Anastasio, M. A., *2D Acoustic Numerical Breast Phantoms for Ultrasound Computed Tomography*. 2025. DOI: 10.13012/B2IDB-5648161_V1.
- [132] **Villa, U.**, Lozenski, L., Park, S., Mert Cam, R., *2D tumor perfusion numerical phantom for dynamic contrast enhanced photoacoustic tomography virtual imaging studies*. Version V1. 2024. DOI: 10.7910/DVN/P7ZVJ0.
- [133] Li, F., **Villa, U.**, *3D Numerical Breast Phantoms and Ring-Array USCT measurements (3 rings)*. Version V1. 2023. DOI: 10.7910/DVN/8JVLAE.
- [134] Park, S., **Villa, U.**, Li, F., Cam, R., Oraevsky, A., Anastasio, M., *3D optoacoustic numerical breast phantoms and simulated OAT measurement data (natural shape, 2 lesions)*. Version V1. 2023. DOI: 10.7910/DVN/OZRVX6.
- [135] Park, S., **Villa, U.**, Li, F., Cam, R., Oraevsky, A., Anastasio, M., *3D optoacoustic numerical breast phantoms and simulated OAT measurement data (hemispherical shape, 4 lesions)*. Version V1. 2023. DOI: 10.7910/DVN/AQZE3H.
- [136] Park, S., **Villa, U.**, Li, F., Cam, R., Oraevsky, A., Anastasio, M., *3D optoacoustic numerical breast phantoms and simulated OAT measurement data (natural shape, 4 lesions)*. Version V1. 2023. DOI: 10.7910/DVN/1ZF00W.
- [137] Lozenski, L., Anastasio, M., **Villa, U.**, *2D numerical mouse phantom for dynamic photoacoustic tomography virtual imaging studies of small animal models*. Version V1. 2022. DOI: 10.7910/DVN/3DXS18.

- [138] Li, F., **Villa, U.**, Park, S., Anastasio, M., *3D Acoustic Numerical Breast Phantoms*. Version DRAFT VERSION. 2021. DOI: 10.7910/DVN/KBYQQ7.
- [139] Li, F., **Villa, U.**, Park, S., Anastasio, M., *2D Acoustic Numerical Breast Phantoms and USCT Measurement Data*. Version V1. 2021. DOI: 10.7910/DVN/CUFVKE.
- [140] Park, S., Brooks, F., **Villa, U.**, Su, R., Anastasio, M., Oraevsky, A., *3D numerical breast phantom: Normalization of optical fluence distribution for 3D functional OAT*. Version V1. 2022. DOI: 10.7910/DVN/1FW2I6.

Open source scientific software

1. Lead developer of hIPPYlib - Inverse Problems Python library (<https://hippylib.github.io>)
2. Lead developer of Elag, ParElag (element agglomeration multigrid solvers and upscaling tools, <http://github.com/LLNL/parelag>)
3. Lead developer of ElagMC, ParElagMC (Multilevel Monte Carlo software based on Elag/ParElag, <https://github.com/LLNL/parelagmc>)
4. Contributor to the finite element library MFEM (<http://mfem.org>)
5. Developer of the finite element library LifeV (www.lifev.org)

CONFERENCE
PRESENTATIONS
AND SEMINARS

Spotlight presentations (Best student paper award)

1. *A Block-Diagonal Algebraic Multigrid Preconditioner for the Brinkman Problem*, 12th Copper Mountain Meeting on Iterative Methods, March 25-30, 2012, Copper Mountain, Colorado, US

Oral presentations at conferences and workshops

2. *A predictive brain cancer digital twin for precision medicine*, Mathematical Challenges in Brain Mechanics Final Conference, June 3-5, 2026, Oslo, Norway (*invited, with travel support for organizers*).
3. *A virtual imaging framework for DCE-PACT estimation of tumor perfusion*, SPIE Medical Imaging, Feb 15–19, 2026, Vancouver, CA
4. *Realistic numerical phantoms and virtual imaging trials to advance quantitative PAT*, Quantitative photoacoustic imaging—from theory to applications, March 18–20, 2025, University of Eastern Finland, Kuopio, Finland (*invited, with partial travel support for organizers*)
5. *Stochastic numerical phantoms to enable optoacoustic tomography virtual imaging studies*, Virtual Imaging Trials in Medicine - International Summit, April 22–24, 2024, Duke University, Durham, NC
6. *Anatomically Realistic Stochastic Numerical Breast Phantoms for Photoacoustic Virtual Imaging Studies and AI-assisted Image Reconstruction*, SIAM Conference on Uncertainty Quantification, Feb 27 - March 1, 2024, Trieste, IT
7. *Low-Rank Matrix Estimation-Based Spatiotemporal Image Reconstruction for 4D Photoacoustic Computed Tomography*, SIAM Conference on Optimization, May 31- June 2, 2023, Seattle, WA, US
8. *Scalable Laplace Approximation for Bayesian Optimal Experimental Design*, 13th International Conference on Monte Carlo Methods, August 16-20, 2021, University of Mannheim, Germany (held virtually)
9. *Curvature Enhanced MCMC Algorithms for Bayesian Inverse Problems Governed by PDEs*, SIAM Conference on Computational Science and Engineering, March 1-5, 2021, Dallas, TX, US, held virtually
10. *Proximal Newton Method for Inverse Problems with Non-smooth Regularization Term*, SIAM Conference on Imaging Science, July 6-9, 2020, Toronto, Canada, held virtually
11. *Bayesian Inference of Fault Properties in Two-phase Porous Media Flow*, 56th Annual Technical Meeting of Society of Engineering Science, October 13-15, 2019, St. Louis, MO, US
12. *Scalable optimal experimental design for large scale non-linear Bayesian inverse problems*, Applied Inverse Problems, July 8-12, 2019, Grenoble, France

13. *hIPPYlib: An Extensible Software Framework for Large-Scale Bayesian Inverse Problems with Quantified Uncertainties*, FEniCS Conference, June 12-14, 2019, Washington, D.C., US
14. *Scalable Methods for Bayesian Optimal Experimental Design Using Laplace Approximation*, SIAM Conference on Computational Science and Engineering, Feb 25- March 1, 2019, Spokane, WA, US
15. *Maximize the Expected Information Gain in Bayesian Experimental Design Problems: a Fast Optimization Algorithm Based on Laplace Approximation and Randomized Eigensolvers*, SIAM Conference on Uncertainty Quantification, April 16-19, 2018, Garden Grove, CA, US
16. *hIPPYlib: An Extensible Software Framework for Large-Scale Deterministic and Linearized Bayesian Inverse Problems*, Texas Applied Mathematics and Engineering Symposium, Sept. 21-23, 2017, Austin, TX, US
17. *Taylor Approximation for PDE-Constrained Optimal Control Problems Under High-Dimensional Uncertainty: Application to a Turbulence Model*, SIAM Conference on Control and its Applications, July 10-12, 2017, Pittsburgh, PA, US
18. *Derivative-informed MCMC for Bayesian Calibration of Stochastic PDE Models*, SIAM Annual Meeting, July 10-14, 2017, Pittsburgh, PA, US
19. *Hessian-based Sampling Techniques for Bayesian Inverse Problems with Stochastic PDE Forward Model*, Applied Inverse Problems, May 29-Jun 2, 2017, Hangzhou, China
20. *Bayesian Calibration of Inadequate Stochastic PDE Models*, SIAM Conference on Computational Science and Engineering, Feb 27-March 3, 2017, Atlanta, GA, US
21. *Bayesian Inverse Problems Governed by Stochastic PDE Models*, Joint Mathematics Meetings, January 4-7, 2017, Atlanta, GA, US
22. *An Analytical Technique for Forward and Inverse Propagation of Uncertainty*, SIAM Conference on Uncertainty Quantification, April 5-8, 2016, Lausanne, Switzerland
23. *Highly Scalable Hierarchical Sampling Algorithms for Gaussian Random Fields*, SIAM Conference on Computational Science and Engineering, March 14-18, 2015, Salt Lake City, UT, US
24. *AMG Solvers for Upscaled Mixed Finite Element Discretizations*, 13th Copper Mountain Conference on Iterative Methods, Apr 6 - 11, 2014, Copper Mountain, CO, US
25. *Multilevel Monte Carlo Simulations with Algebraically Constructed Coarse Spaces*, SIAM Conference on Uncertainty Quantification, March 31 - Apr 3, 2014, Savannah, GA, US
26. *Robust Numerical Methods for the Brinkman Problem*, 9th International Conference on Large-Scale Scientific Computations, June 3-7, 2013, Sozopol, Bulgaria (with travel support from symposium organizers)
27. *Block AMG Preconditioners For Mixed Finite Element Discretization of Porous Media Flow Problems*, 16th Copper Mountain Conference on Multigrid Methods, March 17-22, 2013, Copper Mountain, CO, US
28. *PALADINS: Scalable Time-adaptive Algebraic Splitting and Preconditioners for the Navier-Stokes Equations*, SIAM Conference on Computational Science and Engineering, Feb 25-March 1, 2013, Boston, MA, US
29. *PALADINS: a Scalable Solver for the Navier-Stokes Equations*, SIAM Conference on Parallel Processing for Scientific Computing, Feb 15-17, 2012, Savannah, GA, US
30. *PALADINS: A Parallel Algebraic Adaptive Navier-Stokes Solver*, SIAM Conference on Computational Science and Engineering, Feb 28-March 4, 2011, Reno, NV, US
31. *ALADINS: an ALgebraic ADaptive Incompressible Navier-Stokes solver*, XVIII International Conference on Computational Methods in Water Resources, June 21-24, 2010, Barcelona, Spain (student volunteer with partial travel support)

Poster presentations at conferences and workshops

32. *Integrating Data with Complex Predictive Models under Uncertainty: An Extensible Software Framework for Large-Scale Bayesian Inversion*, NSF CSSI PI Meeting, 2020, Seattle, Wa, US
33. *Systematic Integration of Data with Models under Uncertainty*, 21st Century Imaging Sciences Pathway Annual Retreat, June 7th, 2019, St. Louis, MO, US

34. *Integrating Data with Complex Predictive Models under Uncertainty: An Extensible Software Framework for Large-Scale Bayesian Inversion*, NSF SI2 PI Meeting, 2017, Arlington, VA, US
35. *Hard problems are fine to coarsen*, Computation Postdoc Poster Symposium, March 24th, 2014, Livermore, CA, US
36. *Upscaling Techniques for the Brinkman Problem*, 2013 DOE Applied Mathematics Program meeting, August 6-8, 2013, Albuquerque, NM, US
37. *Towards Scalable Solvers for the Brinkman Problem*, Lawrence Livermore Student Poster Symposium, August 8th, 2012, Livermore, CA, US
38. *Robust numerical methods for the Brinkman problem*, Lawrence Livermore Student Poster Symposium, August 10th, 2011, Livermore, CA, US
39. *ALgebraic time ADaptive splitting schemes for the Incompressible Navier-Stokes equations*, 2011 Georgia Scientific Computing Symposium, Feb. 12th, 2011, Atlanta, GA, US
40. *Multiphysics Multimodel Domain Decomposition: An Application to Conjugate Heat Transfer*, 2010 Georgia Scientific Computing Symposium, Feb. 20th, 2010, Atlanta, GA, US

Seminar presentations

41. *Transcranial Photoacoustic Tomography: Imaging the Brain with Light and Sound*, Acoustics Seminar, The University of Texas at Austin, Austin, TX, April 3rd, 2026, Host *Preston Wilson*
42. *Advancing Emerging Medical Imaging Modalities: The Exciting Promise of Virtual Imaging Trials & AI*, Biomedical Engineering Department, The University of Texas at Austin, Austin, TX, March 25th, 2025
43. *Advancing Breast Ultrasound Computed Tomography: Virtual Imaging Trials and AI*, Acoustics Seminar, The University of Texas at Austin, Austin, TX, March 7th, 2025, Host *Preston Wilson*
44. *Towards Digital Twins of Emerging Medical Imaging Modalities: Mathematical Challenges and Opportunities*, University of Houston, Houston, TX, January 19th, 2024, Host *Annalisa Quaini*
45. *Dynamic Imaging of Tumor Vascular Perfusion using 4D Photoacoustic Computed Tomography*, Scientific Computing Seminars, University of Houston, Houston, TX, April 13th, 2023, Host *Tsornq-Whay Pan*
46. *Dynamic Imaging of Tumor Vascular Perfusion using 4D Photoacoustic Computed Tomography*, Babuška Forum, Oden Institute, Austin, TX, March 10th, 2023, Host *Dingcheng Luo*
47. *Advancing ultrasound and photoacoustic tomography via virtual imaging trials*, Center for Computational Oncology, Oden Institute, Austin, TX, December 7th, 2022, Host *T. Yankeelov*
48. *Three-dimensional stochastic numerical breast phantoms for enabling virtual imaging trials of ultrasound and photoacoustic computed tomography*, Department of Mechanical and Aerospace Engineering, University at Buffalo, Buffalo, NY, November 4th, 2021. Host *D. Faghihi*
49. *Quantitative Photoacoustic Tomography: Inversion Algorithms & Challenges*, Georgia Tech, Atlanta, GA, June 25th-26th, 2019, *1st Annual Photoacoustic & Florescence Tomography Workshop*
50. *Learning from data through the lens of mathematical models: Bayesian Inverse Problems and Uncertainty Quantification*, Department of Mathematics, Emory University, Atlanta, GA, June 24th, 2019. Host *A. Veneziani*
51. *Learning from data through the lens of mathematical models: A gentle introduction to Bayesian Inverse Problems*, Mathematics Department, Washington University, St. Louis, MO, January 28th, 2019. Host *J. McCarthy*
52. *Large Scale Inverse Problems and Uncertainty Quantification: Computational Tools and Imaging Applications*, Electrical & Systems Engineering, Washington University, St. Louis, MO, January 24th, 2019. Host *J. O'Sullivan*
53. *Numerical Upscaling and Multilevel Monte Carlo*, Stanford University, Palo Alto, CA, November 12th, 2014, *Algorithms and Architectures Initiative Annual Meeting*.
54. *Multilevel Monte Carlo Simulations with Algebraically Constructed Coarse Spaces*, Emory University, Atlanta, GA, March 28th, 2014. Host *A. Veneziani*

55. *Towards Scalable Solvers for the Brinkman Problem*, Stanford University, Palo Alto, CA, March 4th, 2014. Host *H. Techelepi*
56. *Numerical Upscaling and Algebraic Multigrid for Mixed Finite Element discretizations*, Lawrence Berkeley National Laboratory, Berkeley, CA, February 11th, 2014. Host *X. S. Li*
57. *Numerical Upscaling and Algebraic Multigrid for Mixed Finite Element discretizations*, Tuft University, Boston, MA, December 6th, 2013. Host *J. Adler*
58. *An Optimal Control Approach for Multiphysics Multimodel Domain Decomposition*, Stanford University, Palo Alto, CA, November 7th, 2013. Host *M. Saunders*
59. *Towards Scalable Solvers for the Brinkman Problem*, Kennesaw State University, Kennesaw, GA, October 5th, 2013. Host *Y. Babenko*

SCHOOLS &
WORKSHOPS BY
INVITATION ONLY

1. *Quantitative photoacoustic imaging—from theory to applications*, March 18-20, 2025, University of Eastern Finland, Kuopio, Finland
2. *Computational Uncertainty Quantification: Mathematical Foundations, Methodology & Data*, May 4-8, 2020, Erwin Schroedinger Institute for Mathematics and Physics (ESI), University of Vienna, Vienna, Austria (virtual)
3. *IdeaLab 2015: Inverse Problems and Uncertainty Quantification*, July 6-10, 2015, Institute for Computational and Experimental Research in Mathematics (ICERM), Brown University, Providence, Rhode Island, Us (with travel support from organizers)
4. *Algebraic Multigrid Summit*, October 15-18, 2014, Boulder, Colorado, US
5. *Algebraic Multigrid Summit*, September 3-8, 2013, Lake City, Colorado, US
6. *Finite Element Exterior Calculus Summer School*, June 11-15, 2012, Institute for Computational and Experimental Research in Mathematics (ICERM), Brown University, Providence, Rhode Island, US (with travel support from organizers)
7. *Adaptive Finite Elements and Domain Decomposition Methods* Workshop, June 17-19, 2010, Milan State University, Milan, Italy

TEACHING
EXPERIENCE

The University of Texas, Austin, TX

Instructor of undergraduate BME courses:

Introduction to numerical methods in BME

Spring 2026

Instructor of Area B courses for the Ph.D. program in Computational Science, Engineering, and Mathematics

Tools & Techniques of Computational Science

Fall 2024

Tools & Techniques of Computational Science

Fall 2023

Tools & Techniques of Computational Science

Fall 2022

Guest lecturer for undergraduate BME courses

Introduction to numerical methods in BME (Dr. Castillo): 1 lecture

Fall 2025

Biomedical Imaging Modalities (Dr. Bush): 1 lecture

Fall 2025

Guest lecturer/Lab-instructor for graduate courses

Computational & Variational Inverse Problems (Dr. Ghattas)

Spring 2025

Computational & Variational Inverse Problems (Dr. Ghattas)

Spring 2024

Computational & Variational Inverse Problems (Dr. Ghattas)

Spring 2023

Washington University, St. Louis, Mo

Instructor of core curriculum courses for the Ph.D. program in Imaging Science

Computational Methods in Imaging Science

Spring 2020

Computational Methods in Imaging Science*

Spring 2019

* *I developed this course.*

Guest lecturer for undergraduate level courses

Optimization (Dr. Kamilov): 2 lectures

Spring 2020

Optimization (Dr. Kamilov): 1 lecture

Spring 2019

The University of Texas, Austin, TX

Instructor of summer schools/short courses

Gene Golub SIAM Summer School on Inverse Problems

June 17-30, 2018

Taught jointly with O. Ghattas, Y. Marzouk, M. Parno, N. Petra, G. Stadler

Lab-instructor for graduate courses

Computational & Variational Inverse Problems (Dr. Ghattas)

Fall 2017

Guest lecturer for graduate level courses

Finite Element Method in Geophysics (Dr. Ghattas): 3 lectures

Fall 2016

Computational & Variational Inverse Problems (Dr. Ghattas): 4 lectures

Fall 2015

Comput. & Variational Inverse Problems (Dr. Petra, UC Merced): 1 lecture

Fall 2015

Emory University, Atlanta, GA

Instructor for undergraduate courses in Calculus I and II

Calculus II (Teaching mentor: Dr. Gould)

Spring 2012

Calculus I (Teaching mentor: Dr. Garibaldi)

Fall 2011

Calculus II (Teaching mentor: Dr. Batterson)

Spring 2011

Teaching Assistant for undergraduate courses in Life Science Calculus and Linear Algebra

Linear Algebra (Lab instructor for Dr. Venapally)

Fall 2012

Life Science Calculus I (Lab instructor for Dr. Duffus)

Fall 2010

Life Science Calculus II (Lab instructor for Dr. Duffus)

Spring 2010

Life Science Calculus I (Lab instructor for Dr. Duffus)

Fall 2009

Life Science Calculus II (Grader for Dr. Duffus)

Spring 2009

Life Science Calculus I (Grader for Dr. Duffus)

Fall 2008

Life Science Calculus II (Grader for Dr. Duffus)

Spring 2008

MENTORING EXPERIENCE

Postdoctoral researchers (main mentor):

1. Jaiqi Yang (2026–)
2. Tao Hong (2025–2026)

Ph.D. students (main advisor):

3. Zeyu Chang (BME, UT Austin, 2026–): Dynamic imaging and neural representations
4. Eric Zou (CSEM, UT Austin, 2026–): Computational wave imaging
5. Victor Couto Kaplan (CSEM, UT Austin, 2026–): Theoretical imaging science—Objective assessment of image quality.
6. Evan Scope Craft (CSEM, UT Austin, 2026): *Data-Driven Optimal Experimental Design and Uncertainty Quantification with Applications in Computational Imaging*
7. Luke Lozenski (Electrical & Systems Engineering, WUSTL, 2025): *AI-assisted Model-guided Image Reconstruction Methods for Photoacoustic and Ultrasound Computed Tomography*

Ph.D. students (co-advisor):

8. Blake Evans (advised by Dr. Castillo, BME, 2026–): Physics-Based Approaches to Quantify Pulmonary Function from Non-Contrast CT
9. Kevin Huang (advised by Dr. Anastasio, Bioengineering, UIUC, 2026): *Advancing photoacoustic tomography neuroimaging through model-based image reconstruction and learning*
10. Refik Cam (advised by Dr. Anastasio, Electrical & Computer Engineering, 2026): *Advanced Image Reconstruction in Photoacoustic Computed Tomography*
11. Gangwon Jeong (advised by Dr. Anastasio, Bioengineering, UIUC, 2025): *Joint image reconstructions methods for solving acoustic inverse problems in ultrasound and photoacoustic computed tomography*
12. Fu Li (advised by Dr. Anastasio, Bioengineering, UIUC, 2024): *Advanced image reconstruction algorithm for 3D accurate and high-resolution breast ultrasound tomography*

Ph.D. students (dissertation committee member, mentor, or research supervisor):

13. Fadeel Sher Khan (**dissertation committee member and co-author**, Biomedical Engineering, UT Austin, advisor: Dr.s Turner and Markey; 2025 –):

14. Kenton Wu (**dissertation committee member**, Computational Science, Engineering, and Mathematics, UT Austin, advisor: Dr.s Dawson and Valseth; 2025 –): *Stabilizing Channel Flow with Local Timestepping and Weak Galerkin Schemes*
15. Panpan Chen (**dissertation committee member and research mentor**, advised by Dr. Anastasio, UIUC, 2022 –): Benchmarking of deep learning methods for photoacoustic computed tomography
16. Kaiyi Yang (**dissertation committee member and research mentor**, advised by Dr. Anastasio, UIUC, 2022 –): Compensation of spatial impulse response in photoacoustic computed tomography.
17. Graham Pash (**dissertation committee member and research mentor**, Computational Science, Engineering, and Mathematics, UT Austin, advisor: Dr. Willcox; 2026): *Towards Predictive digital Twins with Applications in Precision Oncology*
18. Ziheng Zhang (**research mentor**, Computational Science, Engineering, and Mathematics, UT Austin, advisor Dr. Ghattas; 2022 –): Derivative-informed neural surrogates with applications to geophysical problems
19. Brendan Eder (**rotation supervisor**, Biomedical Engineering, UT Austin, Fall 2025): *Principles of photoacoustic imaging formation*
20. Tao Ge (**rotation supervisor**, Electrical & Systems Engineering, WUSTL, Spring 2019): *Proximal Newton Methods for Inverse Problems with Non-Smooth Regularization Term*
21. C. S. Lee (**Summer Internship co-supervisor** with Dr. Vassilevski, LLNL, 2014): Spectral upscaling method for mixed formulation of Darcy equation
22. M. Christensen (**Summer Internship co-supervisor** with Dr. Vassilevski, LLNL, 2013 & 2014): Mixed finite element methods and numerical upscaling with application to subsurface flow and petroleum engineering
23. S. Ladenheim (**Summer Internship co-supervisor** with Dr. Vassilevski, LLNL, 2013): Generation of Gaussian random field by solving stochastic PDEs
24. D. Emerson (**Summer Internship co-supervisor** with Dr. Vassilevski, LLNL, 2013): Non-linear multilevel methods

Ph.D. students (informal mentoring, joint publications):

25. Simone Puel (advised by Dr. Beker, Jackson School of Geosciences, 2023): *A mixed forward/inverse modeling framework for earthquake deformation problems*
26. Tom O’Leary-Roseberry (advised by Dr. Ghattas, Oden Institute, UT Austin, 2020): *Efficient and dimension independent methods for neural network surrogate construction and training*
27. Amal Alghamdi (advised by Dr. Ghattas, Oden Institute, UT Austin, 2020): *Bayesian inverse problems for quasi-static poroelasticity with application to ground water aquifer characterization from geodetic data*

M.S. students:

28. Radhika Bhandari (**Research supervisor**, Mechanical Engineering, 2026–): Primal dual and non-smooth optimization for inverse problems governed by PDEs
29. Jenil Shah (**MS thesis supervisor**, Oden Institute, UT Austin, 2025): *Low Rank-Based Image Reconstruction Methods for Dynamic Contrast Enhanced Multispectral Photoacoustic Tomography*
30. Siva Saket Sripada (**Research mentor**, 2025, Biomedical Engineering): Principles of Photoacoustic Image Formation
31. Venugopal Ranganathan (**MS thesis co-supervisor** with Dr. Ghattas, Oden Institute, UT Austin, 2024): *hIPPYlib: Solving inverse problems in hIPPYlib and FEniCSx*
32. Karan Prakash Hiranandani (**MS report co-supervisor** with Dr. Ghattas, Oden Institute, UT Austin, 2023): *hIPPYfire: Solving inverse problems in hIPPYlib and Firedrake*
33. Joseph Kuo (advised by Dr. Anastasio, Electrical & Computer Engineering, 2022): *Advancing Photoacoustic Neuroimaging Through Deep Learning*
34. Ricardo Qiu (**advisor**, Computer Science Engineering, WUSTL, 2021): *Data-driven approaches to solve inverse problems*
35. Argho Dattas (*Research fellowship mentor*, Electrical & Systems Engineering, WUSTL, Spring 2020): *Learning adversarial regularizers for the solution of inverse problems*

36. Jieqiong Xiao (Research fellowship mentor, Computer Science Engineering, WUSTL, Spring 2020): *ADLA: Automatic differentiation and local assembly of exotic finite element variational forms in MFEM*
37. Di Liu (advised by Dr. Ghattas, Oden Institute, UT Austin, 2017): *hIPPYLearn: An inexact Stochastic Newton-CG method for training neural networks*
38. Ge Gao (advised by Dr. Ghattas, Oden Institute, UT Austin, 2017): *hIPPYLearn: An inexact Newton-CG method for training neural networks with analysis of the Hessian*

Undergraduate students' mentored research:

39. Megan Ondap (**Certificate in Scientific Computation and Data Sciences**, Biomedical Engineering, UT Austin, S2026): *Tomographic Image Reconstruction Using Fourier Neural Operators*
40. Christopher Medrano (**CSE Certificate project**, Biomedical Engineering, UT Austin, S2026): *Optimizing medical image acquisition protocols using task-based measures of image quality*
41. Thomas Wynn (**Undergraduate research assistant**, UT Austin, 2023–2024): *Photoacoustic computed tomography imaging models*
42. Luke Lozenski (**Independent Study Supervisor**, Electrical & Systems Engineering, WUSTL, Summer 2019): *Learning forward modeling error in Photoacoustic tomography reconstruction*
43. Argho Datta (**Independent Study Supervisor**, Biomedical Engineering, WUSTL, Spring 2019): *Proximal Newton Methods for Medical Imaging*
44. Bassel Saleh (advised by Dr. Ghattas):
 Turing Scholars Honors thesis, UT Austin, 2018: *Scientific Machine Learning: A Neural Network-Based Estimator for Forward Uncertainty Quantification*
 Moncrief Undergraduate Summer Internship, UT Austin, 2016: *Neural Networks as Surrogate Models for Forward and Inverse Problems*

SERVICE TO
SCIENTIFIC
COMMUNITY

Editorial work and peer review

Associate editor for IEEE Open Journal of Signal Processing (EiC: Brendt Wohlberg) since 2023.

Editorial board member of Numerical Linear Algebra with Applications (EiC: Panayot Vassilevski) since 2018.

Reviewer for the following journals: *IEEE Transactions on Medical Imaging*, *IEEE Transactions on Computational Imaging*, *IEEE Transactions on Biomedical Engineering*, *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control*, *IEEE Transactions on Image Processing*, *IEEE Transactions on Neural Networks and Learning Systems*, *IEEE Photonics Journal*, *IEEE Journal of Biomedical and Health Informatics*, *IEEE Signal Processing Letters* (IEEE); *Journal of Biomedical Optics* (SPIE); *The Journal of the Acoustical Society of America* (ASA), *SIAM Journal for Uncertainty Quantification*, *SIAM Journal on Scientific Computing*, *SIAM Journal on Imaging Sciences* (SIAM); *Numerical Linear Algebra with Applications*, *International Journal for Numerical Methods in Engineering* (Wiley); *Computational Geosciences*, *Journal of Scientific Computing*, *Numerical Algorithms*, *Advances in Computational Mathematics*, *Numerische Mathematik* (Springer); *Photoacoustics*, *Journal of Mathematical Analysis and Applications*, *SoftwareX* (Elsevier); *Optics Letters* (Optica); *Ultrasonic Imaging* (SAGE); *Journal of Numerical Mathematics* (De Gruyter); *The Journal of Machine Learning for Biomedical Imaging*; *Communications Engineering*, *Scientific Reports* (Nature), *Applied Physics Letters* (AIP)

Grant reviews

Served in 1 NSF grant review panel (Office of Advanced Cyberinfrastructure)

Served as reviewer for the Swiss National Science Foundation (2 proposals)

Served as reviewer for the German Research Foundation (DFG - Deutsche Forschungsgemeinschaft) (2 proposals)

Education and training

Organize and teach the 2018 Gene Golub SIAM Summer School on *Inverse Problems: Systematic Integration of Data with Models under Uncertainty*, in collaboration with O. Ghattas, Y. Marzouk, M. Parno, N. Petra, and G. Stadler

Minisymposia/conference-session organization

1. N. Petra, U. Villa, *Recent Advances on PDE-constrained optimization packages and libraries*, International Conference on Continuous Optimization, July 21-24, 2024, Los Angeles

2. S. Henneking, N. Petra, U. Villa, O. Ghattas, *Computational tools for large-scale inverse problems and UQ*, SIAM Conference on Uncertainty Quantification, Feb 27 – Mar 1, 2024, Trieste, Italy
3. D. Faghihi, K. Maupin, A. Tabarraei, U. Villa, *Data-Enabled Predictive Modeling, Machine Learning, and Uncertainty Quantification in Computational Mechanics*, ASME's International Mechanical Engineering Congress & Exposition, Nov 1-4, 2021, *virtual*
4. D. Faghihi, K. Maupin, A. Tabarraei, U. Villa, *Data-Enabled Predictive Modeling, Machine Learning, and Uncertainty Quantification in Computational Mechanics*, ASME's International Mechanical Engineering Congress & Exposition, Nov 15-19, 2020, *virtual*
5. U. Villa, O. Ghattas, *Optimal Experimental Design for Bayesian Inverse Problems*, SIAM Conference on Computational Science and Engineering, Feb 25-March 1, 2019, Spokane, WA, US
6. U. Villa, T. Oliver, N. Petra, O. Ghattas, R. Moser, *Characterizing model inadequacy in Bayesian inference*, SIAM Conference on Uncertainty Quantification, April 16-19, 2018, Garden Grove, CA, US
7. T. Bui-Thanh, O. Ghattas, V. Rao, U. Villa, *Efficient Algorithms for Bayesian Inverse Problems Governed by PDE Forward Problems*, SIAM Conference on Computational Science and Engineering, Feb 27-March 3, 2017, Atlanta, GA, US

DEPARTMENTAL & UT Austin, Austin, TX

INSTITUTIONAL SERVICE

- BME Graduate Program Admission Committee (203 domestic applicants) **2026**
- CSEM Graduate Program Admission Committee (131 PhD applicants, 39 MS applicants) **2025**
- Ph.D. dissertation committee member: Evans Blake (BME, *current*, Tarini Thiagarajan (CSEM, *current*), Jon Staggs (CSEM, *current*), Kenneth Meyer (CSEM, *current*), Joshua M Bigman (CSEM, *current*), Fadeel Sher Khan (BME, *current*), Eboni Williams (CSEM, *current*), Kenton Wu (CSEM, *current*), Anthony Yu (BME, Georgia Tech, *current*), Graham Pash (CSEM, 2026)¹
- Ph.D. *qualifying exam committee member*: Gutta J. Chowdary (BME, 2026), Holly Zhao (BME, 2026),
- Supervise undergraduate and master research
- M.S. thesis/report reader: Siyi Yu (CSEM, 2026)
- Participate in recruiting events & activities of the Oden Institute and BME Department
- BME Graduate Students Social, “Professor Spotlight”, Oct 2, 2025 **2025**
- Meet & interview candidates for postdoctoral positions

Washington University, St. Louis, MO²

- Member of the Ph.D. in Imaging Science *curriculum committee* (2020-2021)
- Ph.D. *dissertation committee member*: Tingting Wu (IS, 2023), Austen Curcuru (BME, 2023), Shuying Li (BME, 2023), Uri Goldsztejn (BME, 2022), Eghbal Amidi (BME, 2021), Jingwei Lu (ESE, 2019)
- M.S. *thesis committee member*: Shangguan Wentao (ESE, 2021), Weiran Wang (ESE, 2019), Shiqi Xu (ESE, 2019)
- Ph.D. *qualifying exam committee member*: Senyue Hao (ESE, 2022), Tingting Wu (IS, 2020), Zhi Wang (IS, 2020), Soumyendu Ghosh (ESE, 2019), Jiaming Liu (ESE, 2019)
- Supervise undergraduate and master research
- Participate in *recruiting activities* for prospective undergraduate and master students

PROFESSIONAL AFFILIATIONS

SIAM member since 2009.
IEEE member since 2019.
SPIE member since 2026.
BMES member since 2026.

¹BME=Biomedical Engineering, CSEM=Computational Sciences, Engineering, & Mathematics

²BME=Biomedical Engineering, IS=Imaging Science Ph.D. ESE=Electrical & Systems Engineering